

Youssef Khemiri

Data Science and AI Student

📍 Germany — ✉️ khemiri.youssef@esprit.tn — 🌐 /in/youssef-khemiri — 🎧 /youssefkhemiri — 🌐 Portfolio

Summary — International student with over 2 years of experience in developing AI-powered solutions, data engineering, and process automation. Passionate about leveraging emerging technologies and staying at the forefront of industry innovation. Explore my work and initiatives on my **portfolio website**.

Skills

- **Languages:** Python, R, C/C++, Java, JavaScript, PHP
- **Frameworks:** Django, FastAPI, TensorFlow, Scikit-learn, Node.js, Streamlit, Symfony, Springboot, Langchain, DotNet
- **Databases:** SQL(Postgres, MySQL), NoSQL(MongoDB), VectorDB(Milvus, Pinecone, Chroma)
- **Tools:** Git, Docker, AWS, GCP APIs, Hugging Face, MLflow
- **Data Science:** Machine Learning, Deep Learning, NLP, Data engineering, Web Scraping, GenerativeAI
- **Core Strengths:** Problem Solving, Teamwork, Communication, Time Management, Adaptability

Experience

- AI Engineer** — 42RobotsAI (USA – Remote) March 2023 – Present
- Developed AI-driven automation systems to improve decision-making and operational efficiency.
 - Automated large-scale data extraction using web scraping and LLMs; built APIs on AWS EC2 and managed data on AWS RDS.
 - Collaborated with cross-functional teams, and provided on-call support to align AI solutions with business objectives.
- Top-Rated Freelancer** — Upwork Project-Based
- Delivered 14+ successful projects (800+ hours) in Python development, ML, and AI solutions with 100% job success score.
- Machine learning Intern** — Esprit July 2025 – August 2025
- Built a real estate price prediction system using ML, LLMs, and MLOps best practices.
 - Scraped multiple Tunisian real estate websites and enhanced data with description analysis via LLMs.
- AI Engineering Intern** — Arrow International June 2024 – July 2024
- Designed a Smart YouTube Agent using OpenAI models for real-time summaries, key point extraction, and Q&A.
- Python Instructor** — GoMyCode Tunisia May 2023 – June 2023
- Taught Python programming through structured workshops; received excellent student feedback.
- Data Science Intern** — For Right Solutions (FRS) Feb 2023 – July 2023
- Built an end-to-end deep learning platform with Python, Streamlit, and TensorFlow for image classification.
 - Enabled custom model training, transfer learning, and performance evaluation with metrics and visualizations.
- Backend Development Intern** — Booster BC June 2022
- Developed RESTful APIs with Node.js/Express.js and MongoDB, implementing secure authentication with JWT.

Education

- Philipps-Universität Marburg** 2025 - 2026
Exchange Semester - Masters in Data science
- Esprit school of engineering** 2023 - 2026
Engineering degree in Data science and Software developement
- Institute of Applied Sciences and Technology of Sousse** 2020 - 2023
Bachelor in Computer Science

Languages

- **Arabic:** Native
- **English:** Fluent (Professional Proficiency)
- **French:** Fluent (Professional Proficiency)
- **German:** Elementary

Certifications

- Tensorflow for AI, ML and DL - DeepLearning.ai
- Python Development - Upwork
- Generative AI - Upwork
- HIPAA - ProTrainings
- Artificial Intelligence Course - Samsung Innovation Campus
- Applications of AI for Anomaly Detection - Nvidia
- Building Transformer-Based Natural Language Processing Applications - Nvidia

Projects

AI Email Multi-Agent System

- Developed an advanced multi-agent system operating entirely through email communication, powered by Generative AI and NLP techniques.
- Integrated with Google Workspace APIs to handle web research, document analysis with google docs, calendar management, and spreadsheet operations through natural-language email conversations, using a RAG-enhanced agentic architecture.

AI-Meet-Support

- Built an AI agent that processes Google Meet transcriptions to produce comprehensive meeting notes and extract actionable to-do items, with seamless Slack integration for instant sharing.
- Leveraged custom recursive summarization techniques and the LangChain framework for high-quality, structured outputs.

Job Seekers AI Assistant

- Developed an LLM-powered assistant that helps job seekers update and refine their resumes directly in LaTeX using tailored prompts.
- Used Hugging Face open-source models: Llama 3 (3B) locally for CV enhancement, and a fine-tuned Llama 3 (1B) model for realistic interview-question simulation.

Automatic Detection of Pulmonary Nodules

- Conducted research on the latest advancements in medical imaging for pulmonary nodule detection, applied and tested various Deep Learning models for detection, classification and analysis
- Awarded Best Project among Data Science classes in the same category.

Segmentation CNN for Automatic Crater Detection on Mars

- Reproduced research results from the original paper with improved data cleaning pipelines.
- Preprocessed large-scale satellite images (7680×7680 px) and built a U-Net segmentation model for crater detection.

Forecast Cab Booking Demand

- Trained a Machine Learning demand forecasting model achieving an average RMSE of 0.1.
- Performed feature engineering, correlation analysis, and hyperparameter tuning.

INVOLVEMENT

Technical Mentor — lablab.ai	July 2023 – Oct 2023
Member — ARSII	Jan 2022 – Dec 2023
Chairman — IEEE ISSATso Computer society SBC	Jan 2022 – Jan 2023
Ambassador — IEEEExtreme (Competitive Programming Competition)	Julil 2021 – Nov 2021
Member — IEEE	2021 – Present

Interests

Ping-Pong

- If you open my backpack, you'll find two essentials: a laptop for coding, and ping-pong gear for keeping my reflexes just as sharp.